

CommuTiles: Shape-Changeable Modular Computer System Using Proximity Wireless Communication

Masato Goto
The University of Tokyo
Tokyo, Japan
m.goto@mtl.t.u-tokyo.ac.jp

Toshiki Goto
The University of Tokyo
Tokyo, Japan
goto@mtl.t.u-tokyo.ac.jp

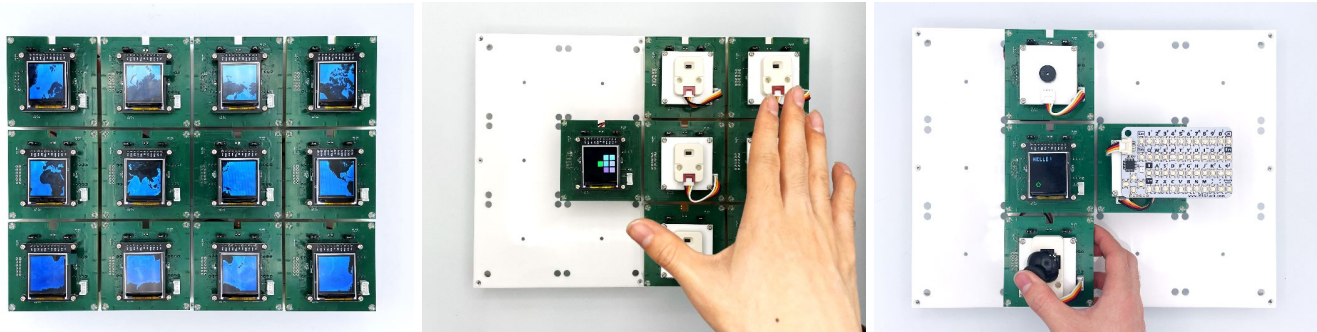
Ibuki Sugiyama
The University of Tokyo
Tokyo, Japan
fuwa@g.ecc.u-tokyo.ac.jp

Junichiro Kadomoto
The University of Tokyo
Tokyo, Japan
kadomoto@mtl.t.u-tokyo.ac.jp

Kenta Higuchi
The University of Tokyo
Tokyo, Japan
higuchi@mtl.t.u-tokyo.ac.jp

Hidetsugu Irie
The University of Tokyo
Tokyo, Japan
irie@mtl.t.u-tokyo.ac.jp

Shuichi Sakai
The University of Tokyo
Tokyo, Japan
sakai@mtl.t.u-tokyo.ac.jp



Map source: Geospatial Information Authority of Japan,
<https://maps.gsi.go.jp/development/ichiran.html>

Figure 1: CommuTiles are composed of a combination of multiple different tile-shaped modules connected wirelessly, forming computer systems of various shapes and functions, and realizing a variety of applications such as a shape-changing display (left), shape-changing sensor (center), and heterogeneous computer system (right).

ABSTRACT

We present CommuTiles, a computer system where multiple tile-shaped modules are wirelessly connected. This represents the first implementation of the concept of a shape-changeable computer system capable of realizing diverse shapes and functions, as well as their dynamic changes. We designed tile-shaped modules equipped with wireless communication circuits and coils, and network protocols for data communication and coordinate estimation. By combining multiple tile-shaped modules with different functions, we implemented application examples. Through the implementation

of hardware, software, and application cases, we lay the foundation for future research on shape-changeable computers and the shape-changing interfaces they enable.

CCS CONCEPTS

• **Human-centered computing** → **Interaction devices.**

KEYWORDS

Shape-changeable computer, shape-changing interface, proximity wireless communication, computing particles, programmable matter

ACM Reference Format:

Masato Goto, Ibuki Sugiyama, Kenta Higuchi, Toshiki Goto, Junichiro Kadomoto, Hidetsugu Irie, and Shuichi Sakai. 2024. CommuTiles: Shape-Changeable Modular Computer System Using Proximity Wireless Communication. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3613905.3648669>

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3648669>

1 INTRODUCTION

As computer systems have penetrated our daily lives with improvements in performance and power consumption, a variety of embedded applications have emerged. In developing computer systems for these applications with diverse implementation forms and functions, it is challenging to meet the shape and functional requirements of each application with a single computer system. Therefore, new developments tailored to specific devices are required, leading to considerable time spent in design and development. One solution to these challenges is to construct a computer system through the combination of multiple standardized computing modules. By assembling various modules as needed, it becomes feasible to easily construct a computer system that satisfies the requirements for different shapes and functions.

Shape-changeable computer system, which allows for the easy construction of computer systems with various shapes and functions through the wireless connection of multiple modules, has been proposed [12]. By equipping each module with a microcontroller and a wireless communication transceiver and connecting them wirelessly, the system as a whole can achieve a variety of shapes, functions, and their dynamic changes. Moreover, by actively exploiting the potential for dynamic shape change, it is possible to construct shape-changing interfaces that use these changes as input/output [1]. Although some basic components have been implemented in hardware prototypes [9, 13, 14], the dynamic shape change of the system and the applications have not yet been demonstrated.

In this paper, we present CommuTiles, a shape-changeable computer system comprised of multiple tile-shaped modules connected wirelessly (Figure 1). Each tile, arranged on a 2-D plane, is wirelessly connected to its adjacent tiles, enabling the system as a whole to achieve a variety of shapes, functions, and their dynamic changes. We demonstrate the hardware and software implementations of CommuTiles and its application cases. This establishes a groundwork for advancing future research and development in the field of shape-changeable computer systems and their varied applications.

2 RELATED WORKS

2.1 Shape-Changeable Computer System

The vision of creating computer systems with various shapes and functions by combining multiple computing modules is not new, and concepts such as Programmable Matter [8, 28] and Robotic Materials [20] have been presented. In recent years, shape-changeable computer systems have been proposed as computer systems that connect multiple modules wirelessly to realize diverse shapes and functions [12]. Implementations of a wireless communication circuit IC [14] and IC equipped with both wireless communication circuits and a processor [13] have been demonstrated. Proposals have also been made for dynamically reconfigurable network protocols between multiple modules [23] and algorithms for estimating the coordinates of each module [9]. Additionally, a programming language targeted at such systems [11], as well as a vision for applying the capability of dynamic shape changes to shape-changing interfaces [15], have also been shown.

CommuTiles, to our knowledge, is the first implementation of the concept of the shape-changeable computer system, composed of

multiple computers that are wirelessly interconnected and capable of dynamically changing its shape and function. In this paper, we present the hardware and software implementation of CommuTiles, along with examples of its applications, and provide insights for future research and development in shape-changeable computer systems.

2.2 Shape-Changing Interface

Shape-changing interfaces composed of multiple modules in tile or block forms achieve various shapes and offer diverse interactions by leveraging their graspable and manipulatable properties [27]. Interfaces composed of multiple modules connected by wired connectors have been proposed previously. Examples have been demonstrated where multiple standardized blocks are connected to estimate various shapes [2] or to realize diverse functions [16]. Furthermore, PickCells [6] have achieved various shapes of touchscreens, while Foxels [24] have realized various shapes of furniture. There are also systems equipped with actuators that support shape changes independently of human intervention. For instance, Robot Pebbles [4] achieve the attachment and detachment of blocks, and 3D M-Blocks [25] implement changes in the position of blocks within the system. These wired communication-based interfaces require wired connectors between blocks. While wired connectors have the advantage of enabling communication between modules and establishing physical connections, they impose constraints on the possible shapes that can be achieved and hinder the smooth transition of the shape.

On the other hand, shape-changing interfaces composed of multiple modules connected wirelessly have also been proposed. The i-Cube [7] consists of cube-shaped modules communicating wirelessly, estimating the arrangement of blocks and the position of each block's surface for various interactions. Shifteo Cubes [21] are made up of multiple modules with displays, estimating and utilizing the position and orientation of adjacent modules. As examples of systems equipped with actuators that change their own shape, there are Kilobots [26]. Here, each module communicates wirelessly, and by combining multiple modules, various shapes are achieved. However, these do not aim for the realization of diverse functions through combinations of modules equipped with different input/output devices, nor do they target the dynamic change of functions.

The CommuTiles proposed in this paper are composed of tile-shaped modules, forming a versatile computer system capable of realizing various shapes, functions, and their dynamic changes. Different input/output devices can be attached with each module, enabling data transmission between modules constituting the system and estimation of coordinates within the system. It becomes a foundational hardware that is universally applicable for the implementation of various applications, including shape-changing interfaces. Each side of the module is equipped with wireless communication circuits, allowing for wireless communication between adjacent tiles. Additionally, a network protocol is implemented as software, enabling data routing. The protocol is designed to be scalable, allowing for the expansion of any number of modules arranged in a 2-D planar grid.

3 DESIGN OF COMMUTILES

3.1 Hardware

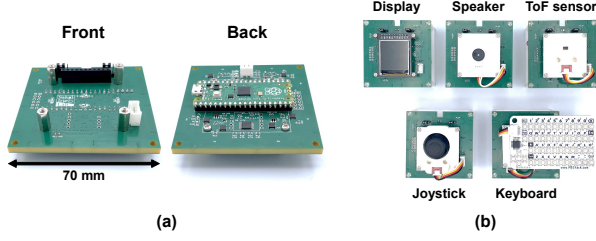


Figure 2: Tile-shaped modules of CommuTiles. (a) The front and back surfaces of the tile. (b) Tiles connected to various devices.

The main components of the tiles that constitute CommuTiles are a microcontroller board, transceiver circuits, and coils for wireless communication. They are equipped with a Raspberry Pi Pico board, utilizing the RP2040 as the microcontroller. For wireless communication transceiver circuits, a high-speed driver IC (Analog Devices ADCLK905) is used as the transmitter, and a high-speed hysteresis comparator IC (Analog Devices ADCMP580) as the receiver, both controlled by the RP2040. The coils are formed using the internal wiring and through-hole vias of the PCB. Each tile is equipped with one transmitting coil and one receiving coil on each side. The size of each tile is 70 mm $\times$ 70 mm, and they can connect with various devices via general-purpose peripheral interfaces such as SPI and I²C (Figure 2a). For the demonstration of multiple applications, tiles equipped with a display (Adafruit ADA2088), speaker (Murata Manufacturing PKM13EPYH4000-A0), ToF (Time of Flight) sensor (M5Stack TOF), joystick (M5Stack Joystick v1.1), and keyboard (M5Stack CardKB v1.1) were prepared, respectively (Figure 2b). Power supply is provided by batteries or wireless powering modules.

For a wireless communication method, proximity wireless communication (PWC) is used. PWC is a technology that has been extensively researched for establishing high-bandwidth connections among semiconductor IC chips [3, 17, 19]. In PWC, the proximity electromagnetic field is utilized, and high-speed base-band pulse signals are transmitted without using a carrier wave. It is possible to transmit data only to inductively coupled coils that are closely placed, and the coils used for communication can be small with a low Q -value. In studies reported thus far, planar spiral coils formed using the internal wiring of semiconductor chips have been used for PWC [22].

In CommuTiles, vertical solenoid coils are formed by utilizing the internal wiring and through-hole vias of the PCB and are used for PWC (Figure 3). This enables the efficient inductive coupling with coils placed on tiles that are horizontally adjacent. Therefore, by placing two small coils on each side of a tile, independent wireless communication channels can be formed, allowing full-duplex communication with each tile that is horizontally adjacent (Figure 4).

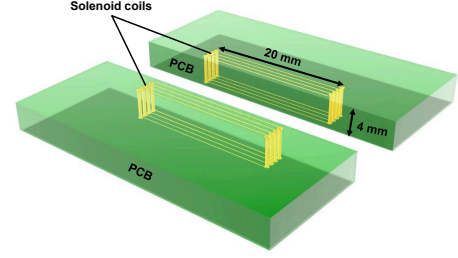


Figure 3: Solenoid coils are formed by using front and back copper layers and through-hole vias of PCBs.

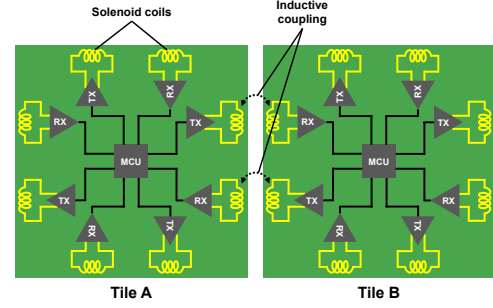


Figure 4: Transceiver circuits and solenoid coils are placed on each side of the tile, forming a full-duplex channel between adjacent tiles. The microcontroller (MCU) is connected to each of the circuits and controls them.

3.2 Software

In CommuTiles, a new network layer protocol is implemented on the microcontroller, which streamlines the process of data transmission and estimation of coordinates between tiles. Every tile is equivalent to a network node, with the system relying on a singular root node for overall coordination. The determination of routing paths and coordinates is reliant on this central root node. Connections between nodes can be configured in a mesh or linear chain arrangement. The protocol we developed effectively creates a virtual tunnel from the root to each node, making the network graph as a star graph. Mesh connectivity allows for multiple potential routes to the root node. We employ the Dijkstra method in our protocol for optimal route calculation. The root node regularly dispatches beacon packets, which are relayed from one node to another. These packets include hop count information, enabling nodes to ascertain the most efficient path to the root node. Moreover, nodes send responses to these beacons, thereby informing the root node of their presence.

4 APPLICATION CASES

The CommuTiles that we have developed are flexible hardware systems capable of supporting a variety of applications. To demonstrate the capabilities of shape and function variability, we have implemented three distinct applications.

4.1 Shape-Changing Display

Tiles equipped with display devices communicate with each other, and by each tile understanding its coordinates within the system, it is possible to realize displays of various shapes. In the implemented application example, a part of the world map is displayed on the display, with the capability of dynamically changing the displayed area (Figure 1 left).

4.2 Shape-Changing Sensor

Similar to the case with shape-changing displays, it is also possible to realize shape-changing sensors. Tiles equipped with sensor devices communicate with each other to acquire sensor data corresponding to each coordinate. In the implemented application example, we visualized the values of the ToF sensors corresponding to each coordinate by varying the intensity of the color displayed on the display device (Figure 1 center).

4.3 Heterogeneous Computer System

By combining tiles with multiple different functions, it is possible to construct a heterogeneous computer system that integrates processors and various input/output devices. In the implemented application example, we have constructed a computer system capable of diverse interactions by combining input devices such as joysticks and keyboards with output devices like displays and speakers (Figure 1 right). It should also be noted that the number of processors and memories can change with the number of tiles, allowing for modifications in the configuration, performance, and power consumption as a parallel computer system.

5 CHALLENGES AND FUTURE DIRECTIONS

The current size of tiles in CommuTiles is 70 mm × 70 mm, and we aim to expand the applicable areas by designing and developing a shape-changeable computer system with smaller tile sizes in the future. It has already been demonstrated that microcontrollers and transceiver circuits can be integrated into a single semiconductor chip measuring 5.3 mm × 2.6 mm [13]. Furthermore, solenoid coils can also be formed on semiconductor chips using technologies such as through-silicon vias (TSV) [18] and a simultaneous vertical and horizontal self-patterning method [10]. Employing these methods to realize semiconductor IC-based tiny tile-shaped modules is our next step.

We will also continue to explore applications and interaction methods possible with shape-changeable computer systems. It has been discussed that the size of each module composing the system could change the appropriate methods of interaction with humans [5]. In the future, by implementing hardware composed of smaller modules and realizing application cases, we aim to advance the evaluation of suitable interaction methods. Furthermore, we plan to progress in research and development not only for systems supporting 2-D grid configurations but also those enabling more diverse arrangements like 3-D, and the applications and interaction methods that utilize these systems.

6 CONCLUSION

This paper presents CommuTiles, a shape-changeable computer system composed of multiple tile-shaped modules connected wirelessly. Each tile, placed on a 2-D plane, is connected to adjacent tiles through proximity wireless communication, enabling the system as a whole to achieve diverse shapes and functions, along with their dynamic alteration. We presented hardware implementation examples of tile-shaped modules equipped with microcontrollers, wireless transceivers, and solenoid coils, as well as network protocol software. Through implemented application cases, we demonstrated the feasibility of constructing computer systems with varying shapes and functions and their dynamic changes. We also discussed the current limitations of CommuTiles and future development directions, providing a foundation for the research and development of future shape-changeable computer systems.

ACKNOWLEDGMENTS

This work was partially supported by Premo Inc. and JSPS KAKENHI Grant Number JP23H00465 and JP23K16857. The authors would like to thank Hirotaka Yamagata for the help with the PCB assembly.

REFERENCES

- [1] Jason Alexander, Anne Roudaut, Jürgen Steimle, Kasper Hornbæk, Miguel Bruns Alonso, Sean Follmer, and Timothy Merritt. 2018. Grand challenges in Shape-changing interface research. *Conference on Human Factors in Computing Systems - Proceedings 2018-April (4 2018)*. <https://doi.org/10.1145/3173574.3173873>
- [2] David Anderson, James L. Frankel, Joe Marks, Darren Leigh, Eddie Sullivan, Jonathan Yedidia, and Kathy Ryall. 1999. Building Virtual Structures with Physical Blocks (UIST '99). Association for Computing Machinery, New York, NY, USA, 71–72. <https://doi.org/10.1145/320719.322587>
- [3] R.J. Drost, R.D. Hopkins, R. Ho, and I.E. Sutherland. 2004. Proximity communication. *IEEE Journal of Solid-State Circuits* 39, 9 (2004), 1529–1535. <https://doi.org/10.1109/JSSC.2004.831448>
- [4] Kyle Gilpin, Ara Knaian, and Daniela Rus. 2010. Robot pebbles: One centimeter modules for programmable matter through self-disassembly. In *2010 IEEE International Conference on Robotics and Automation*. 2485–2492. <https://doi.org/10.1109/ROBOT.2010.5509817>
- [5] Mathieu Le Goc, Lawrence H. Kim, Ali Parsaei, Jean Daniel Fekete, Pierre Dragicevic, and Sean Follmer. 2016. Zoids: Building blocks for swarm user interfaces. *UIST 2016 - Proceedings of the 29th Annual Symposium on User Interface Software and Technology (10 2016)*, 97–109. <https://doi.org/10.1145/2984511.2984547>
- [6] Alix Goguet, Cameron Steer, Andrés Lucero, Laurence Nigay, Deepak Ranjan Sahoo, Céline Coutrix, Anne Roudaut, Sriram Subramanian, Yutaka Tokuda, Timothy Neate, et al. 2019. PickCells: A physically reconfigurable cell-composed touchscreen. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–14. <https://doi.org/10.1145/3290605.3300503>
- [7] Wooi Boon Goh, L. L. Chamara Kasun, Fitriani, Jacquelyn Tan, and Wei Shou. 2012. The i-Cube: design considerations for block-based digital manipulatives and their applications. In *Proceedings of the Designing Interactive Systems Conference (Newcastle Upon Tyne, United Kingdom) (DIS '12)*. Association for Computing Machinery, New York, NY, USA, 398–407. <https://doi.org/10.1145/2317956.2318016>
- [8] S.C. Goldstein, J.D. Campbell, and T.C. Mowry. 2005. Programmable matter. *Computer* 38, 6 (2005), 99–101. <https://doi.org/10.1109/MC.2005.198>
- [9] Toshiki Goto, Junichiro Kadamoto, Hidetsugu Irie, and Shuichi Sakai. in press 2024. Shape Estimation Algorithm for Collective Shape-Changing Interface Using Wirelessly Connected Computers. (in press 2024). <https://doi.org/10.1145/3623509.3635267>
- [10] Kenichiro Hirose, Fumitaka Shiraishi, and Yoshio Mita. 2007. A simultaneous vertical and horizontal self-patterning method for deep three-dimensional microstructures. *Journal of Micromechanics and Microengineering* 17, 7 (jun 2007), S68. <https://doi.org/10.1088/0960-1317/17/7/S02>
- [11] Yusuke Izawa, Junichiro Kadamoto, Hidetsugu Irie, and Shuichi Sakai. 2023. A Functional Reactive Programming Language for Wirelessly Connected Shape-Changeable Chiplet-Based Computers. *SPLASH Companion 2023 - Companion Proceedings of the 2023 ACM SIGPLAN International Conference on Systems, Programming, Languages, and Applications: Software for Humanity (10 2023)*, 61–62. <https://doi.org/10.1145/3618305.3623608>

- [12] Junichiro Kadamoto, Hidetsugu Irie, and Shuichi Sakai. 2019. WiXI: An inter-chip wireless bus interface for shape-changeable chiplet-based computers. *Proceedings - 2019 IEEE International Conference on Computer Design, ICCD 2019* (11 2019), 100–108. <https://doi.org/10.1109/ICCD46524.2019.00021>
- [13] Junichiro Kadamoto, Hidetsugu Irie, and Shuichi Sakai. 2020. Design of Shape-Changeable Chiplet-Based Computers Using an Inductively Coupled Wireless Bus Interface. *Proceedings - IEEE International Conference on Computer Design: VLSI in Computers and Processors* 2020–October (10 2020), 589–596. <https://doi.org/10.1109/ICCD50377.2020.00103>
- [14] Junichiro Kadamoto, Satoshi Mitsuno, Hidetsugu Irie, and Shuichi Sakai. 2020. An Inductively Coupled Wireless Bus for Chiplet-Based Systems. *Proceedings of the Asia and South Pacific Design Automation Conference, ASP-DAC 2020-January* (1 2020), 9–10. <https://doi.org/10.1109/ASP-DAC47756.2020.9045184>
- [15] Junichiro Kadamoto, Takuya Sasatani, Koya Narumi, Naoto Usami, Hidetsugu Irie, Shuichi Sakai, and Yoshihiro Kawahara. 2021. Toward Wirelessly Cooperated Shape-Changing Computing Particles. *IEEE Pervasive Computing* 20 (7 2021), 9–17. Issue 3. <https://doi.org/10.1109/MPRV.2021.3086035>
- [16] Yoshifumi Kitamura, Yuichi Itoh, and Fumio Kishino. 2001. Real-time 3D interaction with ActiveCube. In *CHI'01 extended abstracts on Human factors in computing systems*. 355–356. <https://doi.org/10.1145/634067.634277>
- [17] Atsutake Kosuge and Tadahi Kuroda. 2022. Proximity Wireless Communication Technologies: An Overview and Design Guidelines. *IEEE Transactions on Circuits and Systems I: Regular Papers* 69, 11 (2022), 4317–4330. <https://doi.org/10.1109/TCSI.2022.3210200>
- [18] Harish K. Krishnamurthy, Sheldon Weng, George E. Mathew, Nachiket Desai, Ruchir Saraswat, Krishnan Ravichandran, James W. Tschanz, and Vivek De. 2018. A Digitally Controlled Fully Integrated Voltage Regulator With 3-D-TSV-Based On-Die Solenoid Inductor With a Planar Magnetic Core for 3-D-Stacked Die Applications in 14-nm Tri-Gate CMOS. *IEEE Journal of Solid-State Circuits* 53, 4 (2018), 1038–1048. <https://doi.org/10.1109/JSSC.2017.2773637>
- [19] S.A. Kuhn, M.B. Kleiner, and W. Thewes, R. ISCAS1995 and Weber. 1995. Vertical signal transmission in three-dimensional integrated circuits by capacitive coupling. In *1995 IEEE International Symposium on Circuits and Systems (ISCAS)*, Vol. 1. 37–40 vol.1. <https://doi.org/10.1109/ISCAS.1995.521445>
- [20] M. A. McEvoy and N. Correll. 2015. Materials that couple sensing, actuation, computation, and communication. *Science* 347, 6228 (2015), 1261689. <https://doi.org/10.1126/science.1261689> arXiv:<https://www.science.org/doi/pdf/10.1126/science.1261689>
- [21] David Merrill, Emily Sun, and Jeevan Kalanithi. 2012. Sifteo cubes. *Conference on Human Factors in Computing Systems - Proceedings* (2012), 1015–1018. <https://doi.org/10.1145/2212776.2212374>
- [22] N. Miura, M. Saito, and T. Kuroda. 2012. A 1 TB/s 1 pJ/b 6.4 mm²/TB/s QDR Inductive-Coupling Interface Between 65-nm CMOS Logic and Emulated 100-nm DRAM. *IEEE Journal on Emerging and Selected Topics in Circuits and Systems (JETCAS)* 2, 2 (June 2012), 249–256. <https://doi.org/10.1109/JETCAS.2012.2193836>
- [23] Shun Nagasaki, Junichiro Kadamoto, Hidetsugu Irie, and Shuichi Sakai. 2023. Dynamically Reconfigurable Network Protocol for Shape-Changeable Computer System. *IEEE Design and Test* 40 (12 2023), 18–29. Issue 6. <https://doi.org/10.1109/MDAT.2023.3309891>
- [24] Florian Perteneder, Kathrin Probst, Joanne Leong, Sebastian Gassler, Christian Rendl, Patrick Parzer, Katharina Fluch, Sophie Gahleitner, Sean Follmer, Hideki Koike, and Michael Haller. 2020. Foxels: Build Your Own Smart Furniture. In *Proceedings of the Fourteenth International Conference on Tangible, Embedded, and Embodied Interaction* (Sydney NSW, Australia) (TEI '20). Association for Computing Machinery, New York, NY, USA, 111–122. <https://doi.org/10.1145/3374920.3374935>
- [25] John W. Romanishin, Kyle Gilpin, Sebastian Claici, and Daniela Rus. 2015. 3D M-Blocks: Self-reconfiguring robots capable of locomotion via pivoting in three dimensions. In *2015 IEEE International Conference on Robotics and Automation (ICRA)*. 1925–1932. <https://doi.org/10.1109/ICRA.2015.7139450>
- [26] Michael Rubenstein, Christian Ahler, and Radhika Nagpal. 2012. Kilobot: A low cost scalable robot system for collective behaviors. In *2012 IEEE International Conference on Robotics and Automation*. 3293–3298. <https://doi.org/10.1109/ICRA.2012.6224638>
- [27] Ryo Suzuki. 2019. Collective Shape-Changing Interfaces. In *Adjunct Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology* (New Orleans, LA, USA) (UIST '19 Adjunct). Association for Computing Machinery, New York, NY, USA, 154–157. <https://doi.org/10.1145/3332167.3356877>
- [28] Tommaso Toffoli and Norman Margolus. 1991. Programmable matter: concepts and realization. *Physica. D, Nonlinear phenomena* 47, 1-2 (1991), 263–272.